

Homework 3

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1. Set up

```
library(tidyverse) # general use
library(janitor) # cleaning data frames
library(here) # file/folder organization
library(readxl) # reading .xlsx files
library(flextable) # creating tables

# create kelp object with data
kelp <- read_csv(here("data", "temp-kelp.csv"))
# create bike_commute data with personal data
bike_commute <- read_xlsx(here("data", "ENVS_193DS_Personal_Data_Project.xlsx"))
```

2. Problems

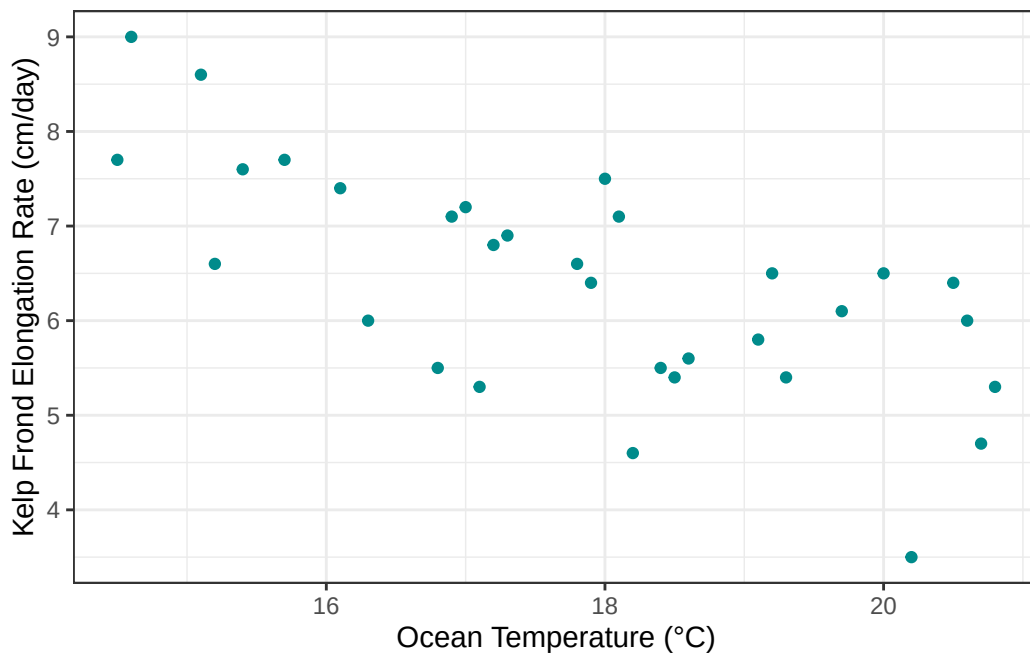
Problem 1. Giant kelp fronds

a. An appropriate test

Because we are interested in the strength of the relationship between temperature and giant kelp frond elongation rate, the appropriate test is a Pearson's r or a Spearman rank correlation. These tests both test for correlation between two variables and see how the variables move together, not necessarily predict each other. Pearson's r test is a parametric test that can be used for linear relationships that have continuous and normally distributed variables. Spearman rank correlation is a non-parametric test, which can be used for monotonic relationships that have interval, ordinal, or nominal data.

b. Create a visualization

```
# base layer: ggplot
# using kelp data frame
ggplot(data = kelp,
       mapping = aes(
         x = temp_c, # x- axis: Ocean temperature (degrees C)
         y = kelp_elong)) + # y axis: kelp elongation rate (cm/day)
# first layer: points
geom_point(
  #changing color from default
  color = "darkcyan") +
# relabeling x- and y- axes
labs(
  x = "Ocean Temperature (°C)", #
  y = "Kelp Frond Elongation Rate (cm/day)") +
# changing theme from default
theme_bw()
```



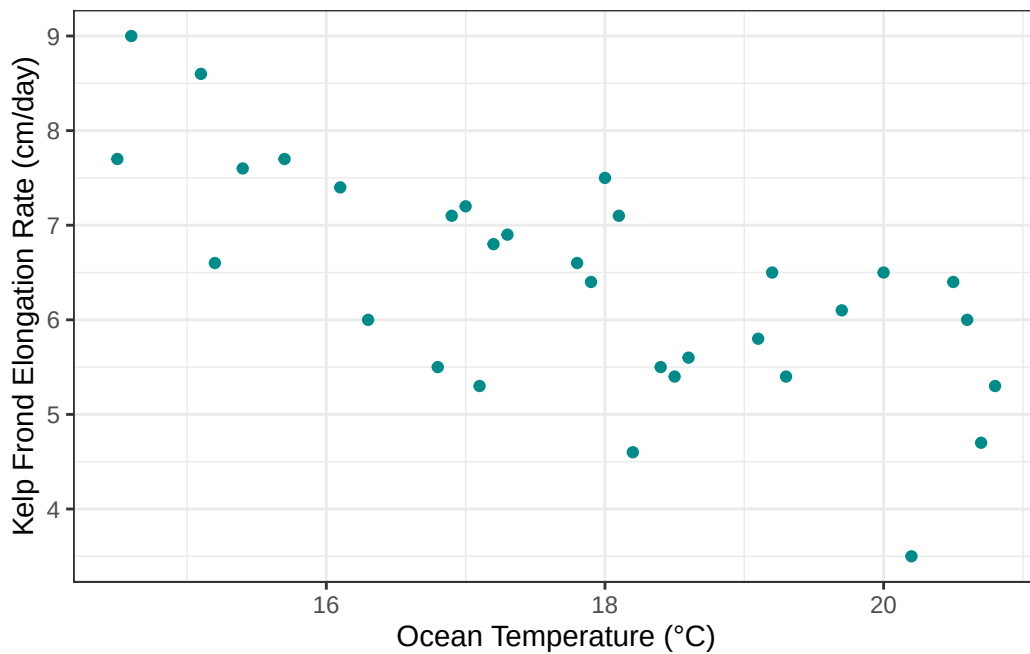
c. Check your assumptions and run your test

Linearity

```

# base layer: ggplot
# using kelp data frame
ggplot(data = kelp,
       mapping = aes(
         x = temp_c, # x- axis: Ocean temperature (degrees C)
         y = kelp_elong)) + # y axis: kelp elongation rate (cm/day)
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```



Model fitting

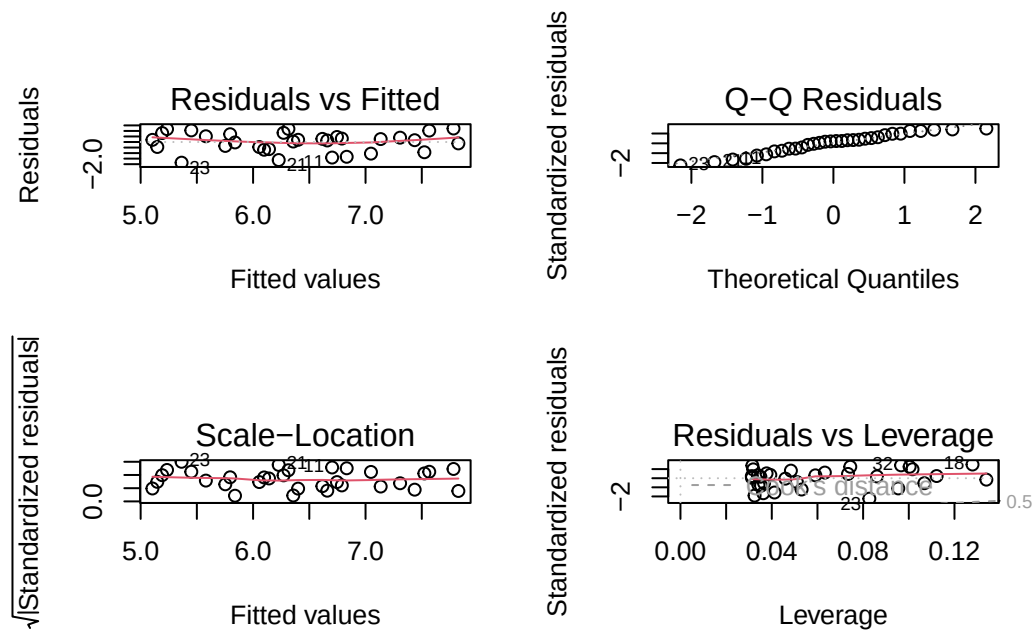
```

kelp_model <- lm(
  kelp_elong ~ temp_c, # formula: change in kelp length as a function of temp
  data = kelp)         # data: kelp data

```

Diagnostics

```
par(mfrow = c(2, 2)) # displays plots in a 2 by 2 grid
plot(kelp_model)      # displays diagnostic plots for model
```



The data is linear and continuous meaning a pearsons r test should be used. In the diagnostic plots, the Q-Q Residuals plot shows that the residuals are normally distributed. The Residuals vs Fitted and Scale-Location plots show that the residuals are evenly distributed with no clear patterns. And finally, the Residuals vs Leverage plot shows that there are no outliers in the data that may have a large influence on the models.

Running test

```
# running correlation test
# variables ocean temperature and giant kelp elongation rate
cor.test(kelp$temp_c, kelp$kelp_elong,
         # running pearson's r based on assumptions
         method = "pearson")
```

Pearson's product-moment correlation

```
data: kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
      cor
-0.6877489
```